

Class Meeting Handout #9 Comments

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Here is the math presented in class today. It is my hope that this document will allow you to follow along with what I am doing on the board and relieve you of the need to write everything that I am doing into your own notes.

This document was produced using the $\LaTeX$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*,

Math is Fun!

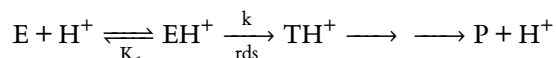
¹ Read chapter 9.3 of the textbook.

Rate laws are just algebra. The trick is to start in the right place.¹

Rate Law for the Specific Acid Catalyzed Hydrolysis

We will begin with acid catalysis via the specific catalysis mechanism. Here, the acid is strong enough to protonate the reactant in a first step. The protonated reactant then goes on to product. In the case of ester hydrolysis the step that follows protonation is the rate-determining attack of the water nucleophile to give the tetrahedral intermediate. Subsequent steps will lead to the products of hydrolysis and return the acidic proton to solution. The acid in specific acid catalysis is the activity of protonation, a_{H^+} , for the system. In dilute water solution, this is equivalent to the concentration of hydronium ion, $[H^+]$.

We can write a scheme for this mechanism featuring equilibrium protonation followed by a rate-determining step.



Scheme 1: Specific acid catalyzed ester hydrolysis. E represents the ester; T the tetrahedral intermediate; and P the eventual products

We could express the rate law as the first-order conversion of the protonated ester to give the tetrahedral intermediate.² Water is the nucleophile, but it is not part of the rate law as it is the solvent and its concentration is not changing.³

$$\frac{dP}{dt} = k[EH^+] \quad (1)$$

Now we need an expression that will give us the value for $[EH^+]$. We can state that EH^+ is an acid and can dissociate according to its K_a equilibrium constant as described by...

$$K_a = \frac{[E][H^+]}{[EH^+]} \quad (2)$$

We do not easily know the values of the concentrations of the protonated ester, $[EH^+]$, or the remaining unprotonated ester, $[E]$. But we do know the total concentration of ester that we added to the experiment...

$$[E]_T = [E] + [EH^+] \quad (3)$$

² Compare my mathematical approach to that of the authors of the textbook and convince yourself that they are both identical. See chapter 9.3.1

³ Later in this course, we might discuss this reaction in extreme acid conditions where there is very little freely available water, and this statement will no longer be accurate. Here, we will use a dilute water solution.

If we can obtain an expression for the concentration of $[EH^+]$ in terms of $[E]_T$ then we would have a value to plug into Eq. 1. We will rearrange Eq. 2 to get an expression for $[E]$ in terms of $[EH^+]$ and $[H^+]$...

$$[E] = \frac{K_a}{[H^+]} [EH^+] \quad (4)$$

and then substitute this into Eq. 3...

$$[E]_T = \frac{K_a}{[H^+]} [EH^+] + [EH^+] \quad (5)$$

Now, we will perform some algebra (collecting terms, adding fractions, etc.) to obtain the expression for $[EH^+]$ that we can substitute into Eq. 1

$$\begin{aligned} [E]_T &= \left(\frac{K_a}{[H^+]} + 1 \right) [EH^+] \\ [E]_T &= \left(\frac{K_a}{[H^+]} + \frac{[H^+]}{[H^+]} \right) [EH^+] \\ [E]_T &= \left(\frac{K_a + [H^+]}{[H^+]} \right) [EH^+] \\ [EH^+] &= \frac{[H^+]}{K_a + [H^+]} [E]_T \end{aligned} \quad (6)$$

Substitute Eq. 6 into Eq. 1 to obtain the rate law...

$$\frac{dP}{dt} = k \frac{[H^+]}{K_a + [H^+]} [E]_T \quad (7)$$

In most examples of specific acid catalysis in dilute acid solution, it is likely that $K_a \gg [H^+]$. For example, the K_a value for most ester groups is greater than 10^6 ($\text{pK}_a -6$) and in dilute acid solution the hydronium ion concentration is less than 10^{-1} ($\text{pH } 1$). So Eq 7 becomes...

$$\frac{dP}{dt} = k \frac{[H^+]}{K_a} [E]_T \quad (8)$$

and we observe the typical first-order dependance on the concentrations of the ester, $[E]_T$, and the hydronium ion, $[H^+]$.

Rate Law for the General Acid-Catalyzed Hydrolysis

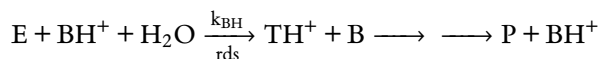
For esters, the specific acid catalyzed pathway is likely to be the major contributor to the overall rate in dilute acid. Hydronium ion has a pK_a value of 0 or -1.74 , depending on who you talk to,⁴ and this is a strong enough acid to transfer the proton to the ester group in a pre-equilibrium step.

However, in cases where the acid is not strong enough for specific acid catalysis, the proton can be transferred in the rate determining step. As the nucleophile attacks, electrons are pushed toward the oxygen of the ester group (in our example), making it more basic. If an acidic group is positioned

⁴ The authors of our textbook use the value $K_a = -1.74$. However, I am increasingly being swayed by the argument for $K_a = 0$. I might actually be able to change my mind after polite debate – who knew? See “The pK_a of Water and the Fundamental Laws Describing Solution Equilibria: An Appeal for a Consistent Thermodynamic Pedagogy.” T.L. Neils, S. Schaertel, T.P. Silverstein, *Helv. Chim. Acta*, 2024, 107, e202400103. <https://doi.org/10.1002/hlca.202400103>

appropriately, the proton can begin to transfer. The act of transferring the proton assists in the nucleophilic attack, and that attack assist the proton transfer. In general catalysis the acid must be a discrete molecule. It could be hydronium or another acid, such as the acidic form of the buffer.⁵

A scheme for general acid catalysis is given below...



and the rate law is stated as...

$$\frac{dP}{dt} = k_{BH}[BH^+][E]_T \quad (9)$$

The water concentration is not included in the rate law because it is solvent. The total ester concentration, $[E]_T$, is used because all of the ester is in the deprotonated form in this mechanism.

The rate depends on the ester concentration and the concentration of the acid. If it is a buffer in water, then there will be a significant portion of the total buffer, $[B]_T$, in both the protonated and unprotonated forms depending the the pH of the solution and the pK_a of the buffer. We can determine the concentration of the acidic buffer, $[BH^+]$ in terms of $[B]_T$ and K_a using the same math as we have already presented above.

$$K_{a_{BH}} = \frac{[B][H^+]}{[BH^+]} \quad (10)$$

and...

$$[B]_T = [B] + [BH^+] \quad (11)$$

We can now perform the same algebra as above to get an expression that describes the concentration of $[BH^+]$. First we will rearrange Eq. 11...

$$[B] = \frac{K_{a_{BH}}}{[H^+]} [BH^+] \quad (12)$$

and then plug Eq. 12 into Eq. 11

$$\begin{aligned} [B]_T &= \frac{K_{a_{BH}}}{[H^+]} [BH^+] + [BH^+] \\ [B]_T &= \left(\frac{K_{a_{BH}}}{[H^+]} + 1 \right) [BH^+] \\ [B]_T &= \left(\frac{K_{a_{BH}}}{[H^+]} + \frac{[H^+]}{[H^+]} \right) [BH^+] \\ [B]_T &= \left(\frac{K_{a_{BH}} + [H^+]}{[H^+]} \right) [BH^+] \\ [BH^+] &= \frac{[H^+]}{K_{a_{BH}} + [H^+]} [B]_T \end{aligned} \quad (13)$$

Substitute Eq. 13 into Eq. 9 to obtain the rate law...

$$\frac{dP}{dt} = k_{BH} \frac{[H^+]}{K_{a_{BH}} + [H^+]} [B]_T [E]_T \quad (14)$$

⁵ Again, compare my mathematical approach for obtaining the rate law in general acid catalysis to that of the authors of the textbook, and convince yourself that they are both identical. See chapter 9.3.2

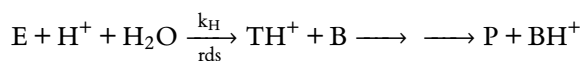
Scheme 2: General acid catalyzed ester hydrolysis. E represents the ester; T the tetrahedral intermediate; and P the eventual products

Observe that we can have a buffered reaction solution set at a give pH value and the rate will increase with first-order dependance on the total concentration of buffer, $[B]_T$.

In general catalysis the rate will increase with buffer concentration. Observe that the rate law for specific catalysis does not include any term for buffer, and the rate should not change with changing buffer concentration in that case.

The Case of General Acid-Catalysis by Hydronium

General acid catalysis can be effected by any acid in solution, including the acidic form of the solvent, H_3O^+ . A scheme for general acid hydrolysis by hydronium is presented below...



and the corresponding rate law can be written as...

$$\frac{dP}{dt} = k_H[H^+][E]_T \quad (15)$$

Observe that the rate has a first-order dependance on $[H^+]$ and $[E]_T$, just like the case of specific acid catalysis.⁶ The fact is thus demonstrated that general acid catalysis by hydronium in water is kinetically identical to specific acid catalysis by hydronium in water. Often, both forms of catalysis by hydronium are declared to be specific catalysis on the erroneous idea that specific acid catalysis is acid catalysis by protonated solvent, in the case hydronium, while general acid catalysis is acid catalysis by buffer. But, the difference between specific and general catalysis lies in the mechanism, not the choice of acid.

Specific catalysis requires that the protonation be an equilibrium preceding the rate determining step. The amount of protonated intermediate will depend only on the acidity of the solution. General acid catalysis involves protonation during the rate determining step. Any acid can be the proton donor: hydronium or buffer or any other acid that is present. Hydronium ion is generally a much stronger acid than the buffer, but the buffer is present at a much higher concentration so we usually see a significant contribution to the overall rate of acid catalysis by the buffer.

Scheme 3: General acid catalyzed ester hydrolysis via hydronium ion. E represents the ester; T the tetrahedral intermediate; and P the eventual products

⁶ This kinetic equivalence is also discussed in your textbook. This document is for helping you follow along with my math during class – there are more details and some useful plots and diagrams in the textbook. See chapter 9.3.3

Determining the Rate Constants

We can take the rate law for specific acid catalysis and design an experiment to obtain the value of the observable rate constant at a given pH, k_{obs} . Consider the rate law as stated in Eq. 8

$$\begin{aligned}\frac{dP}{dt} &= k \frac{[H^+]}{K_a} [E]_T \\ k_{obs} &= k \frac{[H^+]}{K_a}\end{aligned}\quad (16)$$

In a system where we control the pH, we can plot the initial rate of the reaction against $[E]_T$ (when $[E]_T$ is still very close to its initial value). The slope of the line will be k_{obs} . If we know the pH then we can get the value of the k/K_a term. Buffer does not appear in the rate law, so we can use a buffer and it will not affect the kinetics.

In general acid catalyzed systems the rate contribution of buffer catalysis will be significant. We will have general acid catalysis by hydronium (we are in water, after all) and general acid catalysis by the buffer. Each of these two cases are different reactions with different rate constants and the rate laws are given in Eq. 14 and 15. The total rate will be the sum of the rates of all contributing pathways.

$$\begin{aligned}\frac{dP}{dt} &= k_H [H^+] [E]_T + k_{BH} \frac{[H^+]}{K_{a_{BH}} + [H^+]} [B]_T [E]_T \\ \frac{dP}{dt} &= \left(k_H [H^+] + k_{BH} \frac{[H^+]}{K_{a_{BH}} + [H^+]} [B]_T \right) [E]_T\end{aligned}\quad (17)$$

In a buffered situation where pH is unchanging we can define the observable rate constant as...

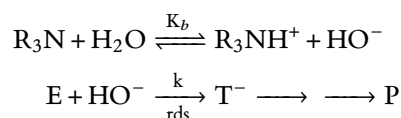
$$k_{obs} = k_H [H^+] + k_{BH} \frac{[H^+]}{K_{a_{BH}} + [H^+]} [B]_T \quad (18)$$

Again, by plotting the initial rate vs $[E]_T$ we can obtain a value for k_{obs} . However, now this value includes a contribution from the buffer rate and k_{obs} will increase if we repeated the experiment using a higher buffer concentration. To properly compare rate constants for acid catalysis we might wish to only report the value for the general acid catalysis by hydronium. We can easily obtain the value of k_{obs} in “zero buffer” by performing the experiment at several buffer concentrations and taking the y-intercept of a plot of k_{obs} vs $[B]_T$. Inspecting Eq. 18 will show you that the y-intercept is the rate for hydronium only and the slope is the value of $k_{BH}[H^+]/(K_{a_{BH}} + [H^+])$.

Rate Law for the Specific Base-Catalyzed Hydrolysis

We may not get to this math in class given the short time of each class meeting. I will present my approach to the rate laws for base catalysis below. You will see that it is the same as the acid catalysis above but we will be bringing in the ion product of water, K_W , to convert hydroxide ion concentration to hydronium ion concentration. The pOH scale is rarely used in physical organic chemistry. We can describe basic solution quite adequately with pH.

In specific base catalysis there is a pre-equilibrium to give a deprotonated intermediate that is then involved in the rate determining step. Imagine adding some triethylamine to our water solution to make it more basic. This base will deprotonate water to give the hydroxide nucleophile that is involved in the rate determining nucleophilic addition to the ester carbonyl group to give the tetrahedral intermediate.⁷ The scheme for this reaction is given below...



The rate depends on the ester concentration and the concentration of hydroxide. We can state the rate law for the rate determining step as...

$$\frac{dP}{dt} = k[\text{HO}^-][\text{E}]_T \quad (19)$$

If we had added sodium hydroxide as the base we would just go with this equation. The rate of reaction would depend on the acidity of the solution as described by pH (remember that we can easily find the hydronium ion concentration in sodium hydroxide solution by $K_W = [\text{H}^+][\text{HO}^-]$). In this case we established that pH value by adding the amine base. So now we must find a term for $[\text{HO}^-]$...

$$\begin{aligned} K_b &= \frac{[\text{R}_3\text{NH}^+][\text{HO}^-]}{[\text{R}_3\text{N}]} \\ [\text{HO}^-] &= K_b \frac{[\text{R}_3\text{N}]}{[\text{R}_3\text{NH}^+]} \end{aligned} \quad (20)$$

and then substitute Eq. 20 into Eq. 19...

$$\frac{dP}{dt} = kK_b \frac{[\text{R}_3\text{N}]}{[\text{R}_3\text{NH}^+]} [\text{E}]_T \quad (21)$$

We know that the acid dissociation constant for the protonated amine base can be expressed as...

$$K_a = \frac{[\text{R}_3\text{N}][\text{H}^+]}{[\text{R}_3\text{NH}^+]} \quad (22)$$

and so...

$$\frac{[\text{R}_3\text{N}]}{[\text{R}_3\text{NH}^+]} = \frac{K_a}{[\text{H}^+]} \quad (23)$$

⁷ This math is also presented in chapter 9.3.1. Compare the textbook method to my own approach and convince yourself that they are identical.

Scheme 4: General base catalyzed ester hydrolysis with an amine as the base. E represents the ester; T the tetrahedral intermediate; and P the eventual products

Substituting Eq. 23 into Eq. 21 gives...⁸

$$\begin{aligned}\frac{dP}{dt} &= kK_b \frac{K_a}{[H^+]} [E]_T \\ \frac{dP}{dt} &= k \frac{K_W}{[H^+]} [E]_T\end{aligned}\quad (24)$$

⁸ Remember that $K_W = K_a K_b$.

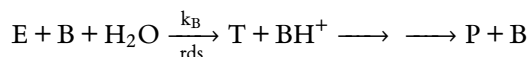
This is the same as Eq. 19.⁹ We have just done a round-about journey to show that specific base catalysis depends only on the basicity of the medium, which can be expressed in terms of acidity using K_W . The rates of both specific acid and specific base catalysis for ester hydrolysis depend only on the acidity of the solution and the concentration of the ester.

⁹ Also, remember that $K_W = [H^+][HO^-]$ or $[HO^-] = \frac{K_W}{[H^+]}$

Rate Law for the General Base-Catalyzed Hydrolysis

In a general base catalysis mechanism the proton transfer occurs during the rate-determining step. In ester hydrolysis this would be represented by a base (hydroxide or the amine buffer, in this case) assisting in water adding to the carbonyl group to form the tetrahedral intermediate.¹⁰

The reaction scheme for general base catalysis could be written as...



¹⁰ You will already have seen Figure 9.8 on page 512 of the textbook when you read chapter 9. This figure shows how a base, buffer or hydroxide, can assist water adding to the carbonyl group. The textbook is a very useful resource.

and the rate law can be written as...

$$\frac{dP}{dt} = k_B [B] [E]_T \quad (25)$$

Scheme 5: General base catalyzed ester hydrolysis. E represents the ester; T the tetrahedral intermediate; and P the eventual products

As always, the concentration of water is included in the value of the rate constant, k_B , because it is constant in a dilute solution.

We will need to get the concentration of the basic form of the buffer, $[B]$ in terms of its total concentration, $[B]_T$. We will do this in the same manner demonstrated in Eqs. 2 through 6 on page 2. We begin with the K_a equilibrium constant for the base as described by...

$$K_a = \frac{[B][H^+]}{[BH^+]} \quad (26)$$

We know the total concentration of buffer that we added to the experiment...

$$[B]_T = [B] + [BH^+] \quad (27)$$

If we can obtain an expression for the concentration of $[B]$ in terms of $[B]_T$ then we would have a value to plug into Eq. 23. We will rearrange Eq. 24 to get an expression for $[BH^+]$ in terms of $[B]$ and $[H^+]$...

$$[BH^+] = [B] \frac{[H^+]}{K_a} \quad (28)$$

and then substitute this into Eq. 25...

$$[B]_T = [B] + [B] \frac{[H^+]}{K_a} \quad (29)$$

Now, we can obtain the expression for $[B]$ that we can substitute into Eq. 25

$$\begin{aligned}
 [B]_T &= \left(\frac{[H^+]}{K_a} + 1 \right) [B] \\
 [B]_T &= \left(\frac{[H^+]}{K_a} + \frac{K_a}{K_a} \right) [B] \\
 [B]_T &= \left(\frac{K_a + [H^+]}{K_a} \right) [B] \\
 [B] &= \frac{K_a}{K_a + [H^+]} [B]_T
 \end{aligned} \tag{30}$$

Substitute Eq. 30 into Eq. 25 to obtain the rate law...

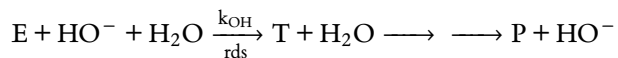
$$\frac{dP}{dt} = k_B \frac{K_a}{K_a + [H^+]} [B]_T [E]_T \tag{31}$$

We have a rate law that displays first order dependence on the total concentration of the buffer, $[B]_T$. Having a contribution to the rate of catalysis by bases other than hydroxide, such as the basic form of the buffer, is a hallmark of general base catalysis.

The Case of General Base-Catalysis by Hydroxide

The strongest base present in water is hydroxide.¹¹ It can also act as a general base.

A scheme for general base catalysis by hydroxide for ester hydrolysis is presented below...



and the corresponding rate law can be written as...

$$\frac{dP}{dt} = k_{OH} [HO^-] [E]_T \tag{32}$$

which can be presented in terms of pH as...

$$\frac{dP}{dt} = k_{OH} \frac{K_W}{[H^+]} [E]_T \tag{33}$$

Compare Eq. 33 with the rate law for specific base catalysis, Eq. 24 on the previous page. Once again we see that general catalysis involving the conjugate base of the solvent is kinetically indistinguishable from specific base catalysis.

The total rate for general base catalysis will be the sum of the rate contributions from all basic species in the system. In this case, the total rate will be the combined rates for general base catalysis by hydroxide ion and by buffer.

¹¹ very strong bases will just produce a stoichiometric equivalence of hydroxide as they are neutralized by water. You will already have read about the levelling effect of solvents on strong acids and bases in chapter 5.2.3 of the textbook.

Scheme 6: General base catalyzed ester hydrolysis via hydroxide ion. E represents the ester; T the tetrahedral intermediate; and P the eventual products

$$\begin{aligned}\frac{dP}{dt} &= k_{OH} \frac{K_W}{[H^+]} [E]_T + k_B \frac{K_a}{K_a + [H^+]} [B]_T [E]_T \\ \frac{dP}{dt} &= \left(k_{OH} \frac{K_W}{[H^+]} + k_B \frac{K_a}{K_a + [H^+]} [B]_T \right) [E]_T\end{aligned}\quad (34)$$

In a buffered situation, where pH is unchanging, we can define the observable rate constant as...

$$k_{obs} = k_{OH} \frac{K_W}{[H^+]} + k_B \frac{K_a}{K_a + [H^+]} [B]_T \quad (35)$$

And, again, we see the opportunity to get the rate constants for both terms by plotting the rate vs $[B]_T$ and a constant pH value.